

E-COMMERCE SALES ANALYSIS AND INTERACTIVE DASHBOARD REPORT

REPORT OVERVIEW

This project involved the analysis of an online retail transaction dataset to evaluate sales performance, customer purchasing behaviour, marketing effectiveness, and order fulfilment efficiency. Microsoft Excel was used as the primary analytical tool to clean the dataset, perform calculations, create PivotTables, and develop an interactive dashboard supported with slicers.

The final solution was presented as a two-page executive dashboard. The first page provides a high-level overview of the business through key performance indicators (KPIs) and operational charts, while the second page focuses on customer behaviour, promotional campaigns, and marketing performance. The dashboard enables decision-makers to interactively explore the data and quickly identify trends that can improve strategic business decisions.

PROJECT OBJECTIVES

The project was undertaken to:

- Analyse overall sales performance.
- Measure customer purchasing behaviour.
- Evaluate coupon campaign effectiveness.
- Assess the contribution of different referral channels.
- Monitor order fulfilment performance.
- Identify high-value customers.
- Build an interactive Excel dashboard for business reporting.

DATASET DESCRIPTION

The dataset consisted of 1,200 customer orders containing the following fields:

- Order ID
- Date
- Customer ID

- Product
- Quantity
- Unit Price
- Shipping Address
- Payment Method
- Order Status
- Tracking Number
- Items in Cart
- Coupon Code
- Referral Source
- Total Price

Each record represented a completed transaction and provided sufficient information to evaluate sales, marketing, customer engagement, and operational performance.

DATA PREPARATION AND CLEANING

Before the analysis commenced, the dataset underwent several preprocessing steps to improve accuracy and consistency.

The following activities were performed:

- Checked for missing and inconsistent records.
- Replaced blank values in the Coupon Code column with "NIL" to distinguish orders that did not use promotional coupons.
- Verified numeric fields such as Quantity, Unit Price, Items in Cart, and Total Price.
- Ensured the Date field was properly formatted for trend analysis.
- Converted the raw dataset into an Excel Table (StoreOrders) to simplify calculations and enable dynamic reporting.
- Built PivotTables to support KPIs and visualizations.

These preprocessing activities ensured that subsequent calculations were reliable and that the dashboard could be refreshed automatically when additional data becomes available.

ANALYTICAL THOUGHT PROCESS

Rather than immediately creating charts, the analysis began by identifying the business questions that management would most likely ask.

These included:

- How much revenue is the business generating?
- How many customers have made purchases?
- Which products contribute the highest sales?
- Which payment methods are preferred by customers?
- Which marketing channels attract the highest revenue?
- Are coupon campaigns effective?
- Who are the most valuable customers?
- How efficient is the order fulfilment process?

Every KPI and visualization included in the dashboard was designed to answer one or more of these questions.

DASHBOARD DEVELOPMENT

The dashboard was intentionally divided into two pages.

Page One: Executive Dashboard

This page provides management with an overview of business performance through KPIs and operational summaries.

KPI Cards

KPI	Value
Total Revenue	1,264,761.96
Total Orders	1,200

Total Customers	1,189
Average Order Value	1,053.97
Order Completion Rate	19.23%
Average Cart Size	5.49 Items

Charts

- Revenue Trend by Date
- Revenue by Product
- Revenue by Payment Method
- Order Status Distribution
- Average Order Value by Payment Method
- Top Best-Selling Products

This page was designed for executives who require a quick understanding of business performance without analysing detailed operational data.

Page Two: Customer and Marketing Dashboard

The second page focused on customer behaviour and promotional effectiveness.

KPI Cards

KPI	Value
Coupon Usage Rate	74.25%
Coupons Used	891 Orders
Average Coupon Order Value	1,057.64
Best Referral Source	Instagram
Top Customer	C38840
Most Popular Payment Method	Credit Card

Charts

- Revenue by Referral Source
- Coupon Performance
- Top Customers by Revenue
- Top Customers by Average Order Value
- Orders by Referral Source
- Revenue by Order Status

Dashboard Interactivity

To improve user experience, slicers were incorporated to allow dynamic filtering by:

- Product
- Payment Method
- Coupon Code
- Referral Source
- Order Status

Using these slicers enables management to instantly analyse specific customer segments, products, or marketing campaigns without modifying the source dataset.

KEY PERFORMANCE ANALYSIS

Revenue Performance

The business generated a total revenue of: \$1,264,761.96

The average order value was approximately: \$1,053.97

This indicates that customers generally placed orders slightly above one thousand currency units per transaction.

Customer Performance

The dataset contained:

- 1,200 orders

- 1,189 unique customers

This suggests that the majority of customers placed only one order during the analysis period, indicating an opportunity to improve customer retention and encourage repeat purchases.

Purchasing Behaviour

Customers purchased an average of:

5.49 items per order

This indicates relatively healthy basket sizes and suggests opportunities to further increase average cart values through product bundling and cross-selling strategies.

Coupon Performance

A total of: 891 orders

used promotional coupons, representing: 74.25% of all orders.

Coupon usage appears to have significantly influenced customer purchases, suggesting that promotional campaigns are an important driver of sales.

The average order value for coupon-assisted purchases was: \$1,057.64

This indicates that customers who used coupons continued to maintain healthy spending levels.

Referral Source Performance

Revenue generated by each referral channel was as follows:

Referral Source	Revenue
Instagram	275,285.45
Email	261,808.55
Google	250,441.48
Facebook	250,410.90
Referral	226,815.58

Instagram generated the highest revenue among all referral sources, indicating that social media marketing is currently the strongest acquisition channel. Referral traffic generated the lowest revenue, suggesting that referral marketing strategies could be improved.

Coupon Distribution

Coupon usage was distributed across four categories:

- FREESHIP
- SAVE10
- WINTER15
- NIL (No Coupon)

The analysis showed that promotional campaigns were widely adopted, with only a minority of customers purchasing without a coupon.

Customer Analysis

The dashboard identified the highest revenue-generating customers.

These customers contributed significantly more revenue than the average customer and should be considered valuable business assets.

Such customers may be ideal candidates for:

- loyalty programmes
- exclusive discounts
- early product launches
- personalised marketing campaigns

Order Fulfilment

The Order Status analysis was designed to evaluate operational efficiency and monitor the progress of customer orders through different fulfilment stages.

The calculated completion rate of **19.23%** suggests that only a portion of all recorded orders were marked as completed (based on the status definition used in the dataset). This metric should be monitored over time to identify bottlenecks in fulfilment and improve customer satisfaction.

Business Insights

The analysis produced several important business insights.

- **Sales:** Revenue is being generated consistently across multiple products and payment methods, demonstrating a diversified sales base.
- **Marketing:** Instagram currently contributes the highest revenue and appears to be the most effective marketing channel.
- **Promotions:** Nearly three-quarters of all purchases involved promotional coupons, highlighting the importance of discount campaigns in driving customer purchases.
- **Customers:** The business serves many unique customers but relatively few repeat purchases, suggesting that customer retention strategies could increase lifetime value.
- **Operations:** Monitoring order completion rates and fulfilment status provides valuable operational insight that can improve delivery efficiency.

RECOMMENDATIONS

Based on the findings, the following recommendations are proposed.

1. **Strengthen Instagram Marketing:** Since Instagram generated the highest revenue, increasing investment in Instagram advertising and content marketing is likely to produce higher returns.
2. **Improve Customer Retention:** Implement loyalty programmes, personalised offers, and reward schemes to encourage repeat purchases among existing customers.
3. **Optimise Coupon Campaigns:** Evaluate which coupon codes generate the highest revenue and continue promoting the most successful campaigns while eliminating less effective promotions.
4. **Improve Referral Marketing:** Referral-generated revenue was comparatively lower. Introducing referral incentives and customer reward programmes may increase referral conversions.
5. **Increase Average Order Value:** Introduce product bundles, volume discounts, and cross-selling opportunities to encourage customers to purchase more items per transaction.
6. **Monitor Order Fulfilment:** Regularly track order status to identify delays and improve operational efficiency, customer satisfaction, and delivery performance.

7. Focus on High-Value Customers: Develop VIP customer programmes for the highest-spending customers through exclusive offers, priority support, and personalised engagement.

CONCLUSION

This project successfully transformed raw transactional data into a comprehensive interactive business intelligence dashboard using Microsoft Excel. Through data cleaning, structured calculations, PivotTables, PivotCharts, slicers, and KPI cards, the dashboard provides management with a clear understanding of sales performance, customer behaviour, marketing effectiveness, and operational efficiency.

The analysis demonstrates that Excel is capable of delivering powerful business intelligence solutions when supported by proper data preparation and thoughtful dashboard design. The completed dashboard serves as a practical decision-support tool, enabling stakeholders to monitor key metrics, identify growth opportunities, and make informed strategic decisions based on reliable analytical evidence.